



Study of the effectiveness of wavelet genetic programming model for water quality analysis in the Uttar Pradesh region

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Abstract Water constitutes an essential part of the earth as it helps in making the environment greener and support life. But water quality and availability are drastically affected by rising water pollution and its poor sanitation. Water gets contaminated due to the excessive use of chemicals by the industries, fertilizers, and pesticides by the farmers. Not only the surface water, groundwater and river water are also getting contaminated. Several published work in Indian context have used different models for the prediction of water quality. Some of them performed poorly due to the presence of irrelevant and missing data in the training samples. Moreover, these studies have assessed water quality on the basis of biochemical oxygen demand (BOD) and coliform and chemical oxygen demand (COD), whereas dissolved oxygen(DO) is one of the most important parameters in terms of water quality assessment as it is considered a key determinant of pollution. Thus, there is a strong need to categorically identify and visualize the DO as one of the key components responsible for deteriorating the

quality of water in Indian context. The main objective of this work is to build a wavelet genetic programming (WGP)-based workflow model for the assessment of water quality in 13 rivers of Uttar Pradesh region. WGP model has a unique feature of discarding the redundant and irrelevant data values from the source data. The proposed WGP model has given promising results which can be attributed to two factors: firstly, the novel use of Morlet wavelet in place of the widely popular Db wavelet, as the mother wavelet, and secondly, the use of MICE technique for missing value imputation in the pre-processing stage. The proposed model not only cleans the data but also demonstrates the feasibility of using DO values as one of the prime factors to assess the water quality.

Keywords Wavelet genetic programming · Dissolved oxygen · Water quality analysis · River water

Introduction

Water is an integral part of life and must be available to all in an easy-to-access manner and in adequate volumes, while being safe to use. Water quality around the globe is rapidly deteriorating, thereby adversely affecting the global ecology. Water pollution is a serious ecological threat caused by various factors such as disposal of sewage into potable and freshwater sources and disposal of industrial wastes into rivers, resulting

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in highly contaminated water. Water contamination is a significant issue in developing countries where the number of potable and freshwater sources is on a constant decline. Approximately 80% of diseases in developing countries are caused due to the usage of contaminated water. As per the World Health Organization, consumption of contaminated water causes around 485,000 diarrheal deaths each year. Using polluted water for irrigation leads to washing away of the essential nutrients from the soil, thus making plants more susceptible to drought and infections.

It is known that for optimum management of water resources along with sustainable growth and development, accurate estimation of water demand needs to be done. But in today's scenario with the increase in industrialization and urbanization, waste production has increased manifolds, and in the absence of an effective and efficient waste disposal mechanism, most of the waste ends up getting dumped into potable and freshwater sources. Furthermore, the excessive use of fertilizers and pesticides by farmers leads to large amounts of fertilizers being present in the soil, getting transferred into the water bodies like rivers and lakes and eventually into the ocean, thus causing environmental damage.

In India, around 70% of river water is contaminated due to the increased concentrations of harmful micro-pollutants. In the same manner, river water reserves are also becoming highly contaminated, gradually causing an alarming situation. Regular assessment of water quality can help to control and reduce the undesirable effects of water pollution. Machine learning (ML)-based approaches have been found to be effective for forecasting values pertaining to water quality parameters. Also, ML-based methods help in obtaining more accurate results faster than traditional methods. At the global level, ML-based approaches like *support vector machine*, *artificial neural network*, and *WGP* (Hadi & Tombul, 2018; Rajaei et al., 2020) are being increasingly utilized for modeling and forecasting water quality. However, in the Indian context, the effectiveness of ML-based approaches for water quality prediction has not been investigated much.

Drawing motivation from the above-mentioned observation, the current work aims at empirically assessing the effectiveness of a hybrid *wavelet genetic programming* (WGP) model for water quality assessment and prediction. The prediction of *dissolved oxygen* (DO) has been considered as the determinant for

assessing the water quality of 13 rivers flowing in the state of Uttar Pradesh (UP). For the proposed work, data was taken from UP government's website for the duration from January 2020 to December 2021.

WGP model was chosen for the current work because:

It helps in computing the solution very fast by discarding the redundant and irrelevant data values that are present in the input variable being used to predict the target variable value.

It has mostly been used in international studies only, and its effectiveness in the Indian scenario remains to be investigated.

Brief literature review

Jamei et al. (2020) have used the previous 20 years TDS data using three different ML techniques, namely, W-MGGP, WGEP, and DWT, from the River Sefid Rud gauging stations Astane and on evaluation based on performance metrics (R, RMSE, NSE) got to know that W-MGGP model was the best model for the prediction of TDS. Ghorbani et al. (2018) devised a procedure to identify how global optimum values of PSR parameters can be used for variation of existing techniques in forecasting the flow of River Araz, whose historical data has been collected from its four gauging stations situated in two different countries, i.e., Iran and Turkey, by using ML models such as LPM, MGGP, PSR, and FNN. So, for the accurate forecasting of river flow, the authors have suggested using the Chaos-MGGP model. It was proved that the Chaos-MGGP model provided more accurate forecasts than the other standalone MGGP or LPM models based on NSE and RMSE. A multi-stage genetic programming–least absolute shrinkage and selection operator (MSGP-LASSO) (Mehr & Gandomi, 2021) to generate a univariate streamflow forecasting model with a 1-month lead time for the data collected from River Sedre, an intermittent river in Antalya, Turkey, has been reported to perform better than other models based on NSE and AIC. Evolutionary programming-based approaches, GEP, FLGGP, and LGP have been used to review the state-of-the-art of GP and its application in water resource system analysis (Azari et al., 2020). The review proved that GP was more effective and efficient with the hybrid variants. Kumar and Sahay (2018) have collected the data from the Kosi

River and Baltara site which consists of details of the historical records of river flow to examine the usefulness of the model for flood forecasting by using ARIMA, DWT, GP, and DWT-GP. It was proved that wavelet transformed data improve the approximation ability of GP. Secondly, the WGP model is the best-performing model. Mehr (2021) collected rainfall data from the Muratpasa meteorology station located in Antalya downtown, Turkey, to increase the accuracy of the seasonal rainfall forecasting. For this, they have created an ensemble evolutionary model that integrates two genetic programming techniques GEP and MSGP. Additionally, they have proposed ensemble models which have increased the seasonal forecasting accuracy of GEP and MSGP models by up to 30%. Finally, GBT was found comparable to the proposed model during the training period. Data collected from Hashtgerd and Zarand-Save sub-catchments located in Salt-Lake catchment, Iran, with various climatic classes in Koppen-Geiger climate classification has been used to estimate the river water quality for agricultural use using various ML models such as Single and integrated-wavelet (integrated-W with noise reduction) soft-computing methods, including ANNs, WANNs, ANFIS, WANNFIS, GEP, and WGEF. Based on RMSE, the WGEF model performed the best (Emadi et al., 2022). Rajaei et al. (2020) reviewed 51 journal papers published from 2000 to 2016 and found out that the various works were dealing with the use of the single and hybrid AI models for river water quality prediction (ANN, GP, FL, SVM, NF, ANN-ARIMA, GA-NN) and wavelet-based hybrid models (WANN, WNF, WSVR and WLGP) out of which they found out that ANN was proved best. Secondly, combining this model with wavelet transform, genetic algorithm, fuzzy logic, and ARIMA has improved the accuracy of water quality prediction. Data was collected from Diga and Jasenska sub-areas of river Neretva coastal aquifer system to the main mechanism influencing the water level, EC, and temperature in ground and surface water, using two different ML models, namely, CWT and WTC (Lovrinovic et al., 2022). Results proved that the wavelet-based analysis enabled the determination of the interdependence of the variables. Baigang et al. (2021) have collected data from waterworks located in Suzhou, China, to understand how water demand can be forecasted using various hybrid ml models which are proposed such as DWT-LSTM,

PCA-LSTM, DWT-PCA-LSTM, DWT-PCA-RNNs, DWT-PCA-BP, and DWT-PCA-SVM. Based on evaluation metrics, namely, pMAPE, MAPE, EVS, and R, the model DWT-PCA-LSTM was found to outperform. Data was collected from five hydrologic stations on the Lower Minnesota River, which is the main tributary of the Mississippi River, USA, to study whether the water quality parameters can be analyzed using statistical and wavelet analysis (Gao et al., 2018).

Altunkaynak and Nigussie (2017) have collected 23 years of monthly water consumption data from January 1992 to December 2014 from the Istanbul Metropolitan Municipality Water Works and Canalization Administration (ISKI) to understand whether the monthly water can be predicted using wavelet-based models such as DWT, MSA, MLP(ANN), DWT-MLP, and MSA-MLP. It was proved that MSA is a better pre-processing technique than DWT. Data collected by Liu et al. (2021) from Lake Ebinger wetland national reserve was used to understand how total nitrogen can be predicted for managing the pollution using models such as DWT and FOD with ML models (RF, Bagging). It has been reported that the XGBoost model with DWT coupled with gray relation analysis (GRA) yielded the best estimation accuracy. Poul et al. (2019) collected samples from River St. Clair, central-northern America, to predict the monthly stream using different ML models such as MLR, ANN, ANFIS, and KNN with wavelet analysis and found which proved that the performances of three nonlinear models ANN, ANFIS, and KNN were highly satisfying. The model which benefits from the advantages of both fuzzy inference systems and neural networks was superior. Seo et al. (2017) have used models such as DWT, MODWT, MLP, SVM, and GA to understand how the daily river stages can be modeled for the data that is been collected from 4 stream-flow gauging stations. Based on the results obtained, MODWT-SVM model has been reported to achieve better efficiency and effectiveness in comparison to both the single and hybrid models created. It was also found that hybrid models coupling MODWT, SVM, and GA can enhance model performance and accuracy in daily river stage modeling as compared with those combined with DWT. Further, the authors also created a MODWT-SVM model using a c12 mother wavelet and based on efficiency indices. It was found that the proposed model can be used in model

efficiency and effectiveness in daily stage river modeling. Supervised ML models (decision tree, artificial neural network, Bayesian network, and genetic programming) and an unsupervised ML model (wavelet transform) have been used to predict the BOD for the Karaj River of Iran (Jafari et al. (2020)). The obtained results signify that WGP technique is the best, but due to less knowledge and approach, the authors were not able to establish more on it. Ravansalar et al. (2017) have collected data from two gauging stations on River Besah, to predict the monthly streamflow using four ML models, namely, single LGP, WANN, ANN, and MLR. The authors proposed a hybrid WLGP model by combining LGP and DWT, and based on performance metrics (RMSE, MAE, and E), it was observed that the proposed model was best for modeling the streamflow. Six important water quality parameters, namely, COD, NH_4^+ -N, DO, EC, WT, pH, and TU, were analyzed by Huang et al. (2018), to predict the River Pearl water quality which is situated in China using FNN, WNN, and NN models. FWNN model was proposed using four methods such as FL, NN, GA, and WT to simulate the water quality prediction and parameters' nonlinearity. A self-adapted fuzzy C-means clustering was used to determine the number of fuzzy rules. Therefore, it was proved that FWNN performed better than the other models. Hadi and Tombul (2018) have analyzed a case study of a basin located in southeast Turkey to understand how streamflow can be forecasted. For this, they have used three models, namely, CWT, ANN, and DDM. After analysis of CWT, important scales were selected to impose into ANN. Before modeling, WCT analysis was conducted to analyze the relationship between variables in the time–frequency domains. It was used to prove that the proposed model can be used to forecast the streamflow prediction. Wu and Wang (2022) collected data from River Jinjiang to predict water quality using multiple ML models, namely, ANN, DWT, LSTM, NAR, and CNN-LSTM. The authors also proposed a hybrid model ANN-WT-LSTM which proved to be more efficient than the other models based on performance metrics MSE, RMSE, MAE, and MAPE. The SARIMA model was used by Martínez-Acosta et al. (2020) to predict the monthly rainfall time series. Data was collected from 75 stations on the River Sinu, Colombia, and it was shown that the SARIMA model can be used for various water resource management and hydrologic management

studies. Tadesse and Dinka (2017) also used the SARIMA model to forecast the mean monthly River Waterval flow which is situated in South Africa. For this, they have forecasted the river flow using the SARIMA model in the software GRET. Based on performance metrics AI, SB, HQ, RMSE, and MAE, it was proved that for river flow prediction, this model can be used. Performance comparison of SARIMA and ANN models based on statistical performance has been done by analyzing the four water quality parameters (COD, DO, pH, $\text{NH}_3\text{-N}$) which are collected from 8 monitoring stations (Wang et al., 2020). Based on MSE, it was established that optimal SARIMA is inferior to the ANN model. Azad et al. (2022) have collected daily reservoir water level data from Red Hills reservoir, Tamil Nadu, to predict the water quality using SARIMA, ANN. These authors have proposed a hybrid SARIMA-ANN model, and based on performance metrics such as RMSE, MAE, MAPE, and R^2 , it was proved that the proposed model performed much better than the standalone models and can be used for reservoir prediction. Momeni and Bonakdari (2016) collected 30 years of monthly data flow for the River Jamishan to forecast the monthly flow varying with an extreme seasonal variation using SARIMA and ANN models. The authors proposed a hybrid SARIMA-ANN model, and based on performance metrics RMSE and the coefficient of determination, it was shown that the proposed model is more efficient than the standalone models. A review of missing data imputation techniques has been done by Mirzaei et al. (2021). It was demonstrated that through missing value imputations, we can mitigate the missing data occurrence. In their work to understand whether unsupervised ML methods can be used to handle missing values, Raja and Thangavel (2019) have used *K*-means, fuzzy *C*-means, and rough *K*-means imputation algorithms (centroid-based and parameter-based imputation techniques). The authors have worked on data taken from the UCI repository, and based on RMSE and MAE, it was proved that the rough *K*-means centroid-based method and *K*-means centroid-based method was proposed by the authors which proved that both the models performed best than the other models. The use of ANN and MICE techniques to determine the missing value imputation has been demonstrated by Tripathi et al. (2022). The authors have proposed a hybrid model using ANN and MICE called extended ANN which proved to be

more efficient than the other models under consideration. Data of 2190 samples collected from the River Canal in the Yangtze River Delta of Jiangnan, China, as used to determine the water quality by analyzing the DO parameters using RF, ABM, and GBU models (Huan et al., 2022). Based on MAE, RMSE, and R^2 , attention-based GRU model was proved best as it outperformed the other models. Marcelino et al. (2022) have used the KNN technique to understand whether the missing data values may affect the regression results. But it was proved that KNN can work better than the other models in terms of regression which are having missing values. Table 1 presents a summary of the studied literature.

Research gaps

For the current work, the following research gaps were identified by going through the existing literature papers:

None of the current published work by Indians authors has accessed the water quality based on the WGP model for assessment.

In terms of water quality assessment, biochemical oxygen demand (BOD), coliform, and chemical oxygen demand (COD) are the only used predictions in the Indian research community. So, the current work focuses on analyzing DO in terms of river water quality assessment.

DO is one of the most important parameters in terms of water quality assessment as it is considered a key determinant of pollution.

Most authors have used ANN or RF techniques for missing value imputations. But the proposed workflow used multivariate imputation by chained equations (MICE) as it helps us to estimate the best prediction for each missing value to get an accurate result for our project.

So far, only Db wavelets have been used but the current work uses the Morlet wavelet as the mother wavelet.

The main objective of the presented work is to study the effectiveness of the WGP model for the assessment of water quality in the UP region. Along with studying the effectiveness of the WGP model, it also explores whether the DO future values are high or low, based on which the water quality level (fit/unfit) can be assessed for usage.

Materials and methods

Wavelet genetic programming

Wavelet transform is a very efficient technique for time–frequency analysis. It analyzes and evaluates noisy signals for the multi-resolution processing of signals with the help of certain mother wavelets. Mother wavelets are nothing but transformation functions. Mother wavelet can be scaled suitably and can be shifted along the original time series, enabling the representation of time series in the time–frequency domain instantaneously and thus is suitable for analyzing both stationary and nonstationary time series. Mother wavelets are of two types, i.e., discrete wavelet transforms (DWT) and continuous wavelet transform (CWT). CWT is suitable for representing a time series in time–frequency domain, while DWT is useful for time series compressing. Based on the current context of water quality assessment, CWT is considered to be more relevant than DWT. Because of this factor, CWT was chosen as the wavelet transform technique. Secondly, genetic programming was used, as it acts as a potential tool to develop efficient and simple functional relationships between hydrological variables.

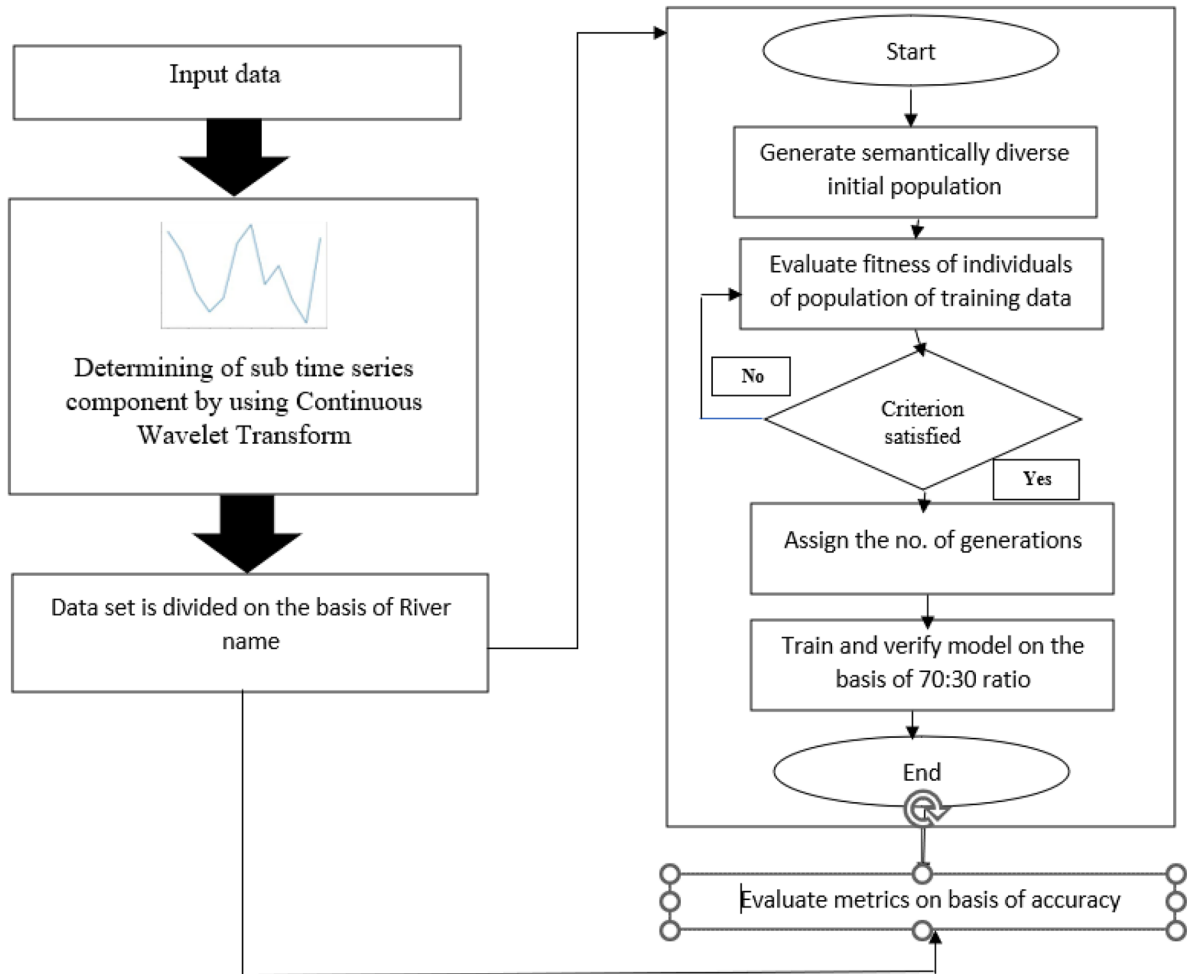
The WGP model is created basically as a combination of continuous wavelet transform strength and processing of genetic programming to achieve the best solution for the water quality assessment. In this model, DO values are decomposed and provided as input variables, through wavelet transform technique, to the genetic programming model to study the effectiveness of the model for the river water quality assessment. In the analysis, CWT was utilized to decompose the river dataset into various resolution scales, which can recognize the active and prominent decomposed sub-series in the DO signals. The original DO signal was decomposed into five-levels. Based on research, it was found out that most used wavelets are db4 (Emadi et al., 2022; Jamei et al., 2020; Ravansalar et al., 2017), followed by db5 (Jafari et al., 2020; Wu & Wang, 2022) and dmey (Jafari et al., 2020; Jamei et al., 2020). The third most used wavelets are bior6.8 (Jamei et al., 2020), Meyer (Rajaei et al., 2020), Morlet (Lovrinovic et al., 2022), db (Gao et al., 2018), and coif (Gao et al., 2018). Liu et al. (2021) used coif, sym, and dbN. Seo et al. (201) have used c12 wavelets. Lastly, Jafari et al. (2020) used db2, bior1.1, coif1, and Haar wavelets. Morlet

Table 1 Sampling locations with the ML model(s), prediction parameter(s), and the performance metrics

Paper reference	Sampling water body	International(I)/national (N)	ML model(s) used	Parameter predicted	Performance metrics
Jamei et al. (2020)	River Sefid Rud	International	DWT, GEP, W-MGGP, WGEP, MGGP	River flow, TDS	RMSE, NSE, <i>R</i>
Ghorbani et al. (2018)	Black Sea	International	W-MGGP, FNN	River flow	RMSE, NSE
Mehr and Gandomi (2021)	River Sedre	International	GP, W-MGGP, MSGP	Streamflow	RMSE, NSE, AIC _C
Kumar and Sahay (2018)	River Kosi	National	DWT, GP, WGP	River flow	RMSE, NSE
Mehr (2021)	Sea	International	GEP, MSGP, MLR	Rainfall	RMSE, NSE, MAPE
Emadi et al. (2022)	Salt Lake	International	ANN, GEP, WANN, ANFIS, WGEP	RD, RW, CA, OA	RMSE, MAE, <i>R</i>
Rajaei et al. (2020)	Various rivers	International, national	SVM, FL, WLGP	DO, temperature, BOD, EC, pH, TDS, algae, SSL, SSC, salinity	RMSE, MAE, <i>R</i> ² , AARE, <i>d</i> , CE
Lovrinovic et al. (2022)	River Neretva	International	CWT	Temperature, EC	Evaluated time series only
Baigang et al. (2021)	Real waterworks	International	DWT, LSTM	Temperature, water demand	MAPE, <i>R</i>
Gao et al. (2018)	River Minnesota	International	NO ₃ , NH ₄ , PO ₄ , Chlorophyll, TSS	DO, temperature, BOD, TP, TKN	<i>R</i>
Altunkaynak and Nigussie (2017)	ISKI	International	DWT, MLP	Water demand	RMSE, CE
Liu et al. (2021)	Lake Ebinur	International	DWT	Temperature, rainfall	RMSE, <i>R</i> ²
Poul et al. (2019)	River St. Clair	National	ANN, ANFIS, MLR	TKN	RMSE, NSE, MAE, <i>R</i>
Seo et al. (2017)	Rivers & Lakes	International	SVM, MLP	Water demand	RMSE, MAE, <i>R</i> ² , AARE, <i>d</i> , CE
Jafari et al. (2020)	River Karaj	International	GP, WANN, WGP	BOD	RMSE, MAE
Ravansalar et al. (2017)	River Beshar	International	ANN, WANN, WLGP	River flow	RMSE, MAE
Huang et al. (2018)	River Pearl	International	ANN, FNN, FL	DO, EC, pH, COD, NH ₄ ⁺ -N, WT, TU	RMSE, <i>R</i> ² , MAPE, <i>R</i> , MSE
Hadi and Tombul (2018)	River Goksu	International	ANN, GP, MGGP, CWT	Temperature, rainfall, streamflow, Eva transpiration	RMSE, <i>R</i> ²
Wu and Wang (2022)	River Jinjiang	International	ANN, DWT, LSTM	DO, COD, TP, NH ₃ -N	RMSE, MAE, MAPE, MSE
Martínez-Acosta et al. (2020)	River Sinu	International	SARIMA	Rainfall	AIC _C
Tadesse and Dinka (2017)	River Waterval	International	SARIMA	River flow	RMSE, MAE
Wang et al. (2020)	CNEMC	International	ANN, SARIMA	DO, pH, COD, NH ₃ -N	RMSE, NSE, MAPE
Azad et al. (2022)	Lake Puzhal	National	ANN, SARIMA	Water demand	RMSE, MAE, <i>R</i> ² , MAPE
Momeni and Bonakdari (2016)	River Jamishan	International	ANN, SARIMA	River flow	RMSE, <i>R</i> ² , bias, MARE

Table 1 (continued)

Paper reference	Sampling water body	International(I)/national (N)	ML model(s) used	Parameter predicted	Performance metrics
Huan et al. (2022)	Yangtze River Delta	International	RF, ABM, GBU	DO	RMSE, MAE, R^2

**Fig. 1** The proposed workflow for WGP model

wavelet is the mother wavelet used in the presented analysis as it helps in producing more accurate results than the Fourier transform and then divides our data on the basis of 13 river names.

WGP model helps in computing the solution very fast by discarding the redundant and irrelevant data that are present in the input variable which is used to predict the target variable value. WGP model is considered the best and superior in terms of other models as it requires less information and gives the best

global solution. The proposed workflow of the model based on the existing WGP model is shown in Fig. 1.

According to the proposed workflow, firstly, input data goes through CWT for the analysis of time–frequency information and helps in the building of classical short-time Fourier transform. Thereafter, the input variables were generated, which explicitly capture the DO values in context for the different water parameters. Then the fitness of the training data values of the individual population is based on various

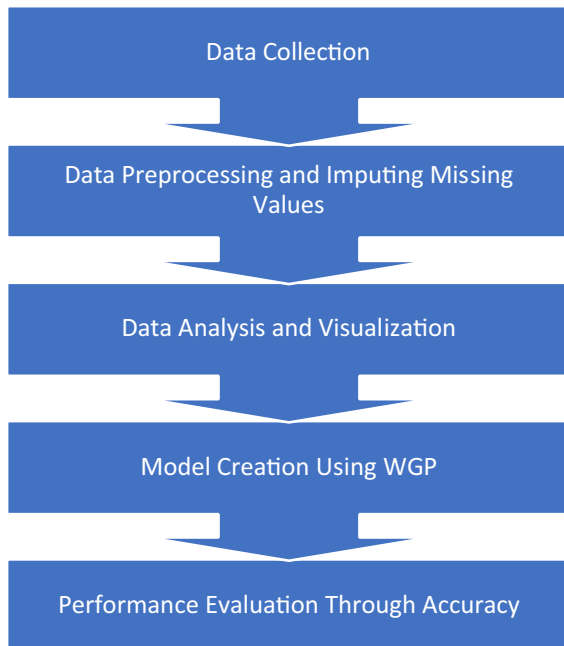


Fig. 2 Methodology used

parameters as evaluated. After that, it was checked whether the criteria requirement, as per the input and output, is satisfied or not. If the criteria get satisfied, then a new number of generations were assigned as parameters, and according to it, the model was trained and validated on the basis of 70:30 ratio. Lastly, the accuracy of the model was used as a metric to evaluate the model's performance.

Proposed workflow

Figure 2 depicts the methodology followed for carrying out the presented work.

The first step was data collection. Data was downloaded from the Uttar Pradesh Pollution Control Board website (<http://www.uppcb.com/water-quality-data-stretches.htm>) which contained the data of 13 different rivers collected from various sampling locations present in the different regions of UP for the years 2020 and 2021.

Data pre-processing

Pre-processing of the data was done to make data more understandable. The following changes were done:

The data type of year and month column was changed. A new DateTime column was created based on year, month, and day, and then the year, month, and day columns were deleted and the DateTime column indexing was done.

Some columns which were not required were removed.

The processed data set was then ready for the next step.

Missing value imputations

It is a technique in which the missing data gets replaced with some other value that can be substituted to retain and acquire most of the dataset information. Null values present in the processed dataset were filled by the missing value imputations technique using the advanced MICE technique. It is a robust, informative method of dealing with missing data in datasets. The procedure “fills in” (imputes) the missing data in a dataset through an iterative series of predictive models. In each iteration, each specified variable in the dataset is imputed using the other variables in the dataset. These iterations must be run until it appears that convergence has been met.

Based on the literature survey, it was found that to get an accurate result, some researchers have used missing data imputation techniques. The most used imputation techniques are based on artificial neural network ANN and random forest (Huan et al., 2022; Marcelino et al., 2022; Tripathi et al., 2022). Mirzaei et al. (2021) have used MCAR, MAR, and MNAR as missing data imputation techniques. Raja and Thangavel (2019) have used various missing value imputation techniques based mainly on *K*-means, fuzzy *C*-means, rough *K*-means, *K*-means centroid-based imputation algorithm, fuzzy *c*-means centroid-based imputation algorithm, *K*-means parameter-based imputation algorithm, fuzzy *c*-means parameter-based imputation algorithm, rough *K*-means centroid-based imputation algorithm, and rough *K*-means parameter-based imputation algorithm. MICE and extended MICE were used by Tripathi et al. (2022) for missing value imputations. Huan et al. (2022) have also used an attention-based mechanism-gated recurrent unit, ANFIS, RBF-ANN, ELM, and SVR. Lastly, DT, KNN, SVM, NN, and AdaBoost have been used by Marcelino et al. (2022).

For missing value imputation, the following steps were taken:

Missing values were detected either numerically or visually.

Then, the percentage of the missing values for each dataset column was detected, to obtain an idea about the missing value distribution.

Thereafter, missing values of DO, BOD, COD, total coliform, and fecal coliform were identified, and the reason for these missing values was explored.

Further, it was observed that the DO column is somehow correlated with the other parameters such as BOD, COD, and coliform (total + fecal); hence, it can be said that missingness in this dataset is a case of missing at random.

Finally, we have treated missing values based on MICE technique, as it helps to estimate the best prediction for each missing value, thereby assisting in getting a more accurate result.

After imputation of missing values, the below mentioned data pre-processing was done:

The column DO (mg/L) was renamed to DO.

BOD, COD, total coliform, fecal coliform, and category columns were dropped.

Indexing was done on the DateTime column, and the previous DateTime column was dropped.

Sorting was performed on the index DateTime column.

Data analysis and visualization

Data analysis

Data analysis is a technique used in data mining for the raw data transformation in a very useful, efficient manner. To get some deeper knowledge, the data was analyzed in the following ways:

The shape of the dataset is observed.

The summary of the dataset was checked.

Columns were verified to be present in the dataset based on their name.

Rows not required were dropped.

The type of data as validated.

Table 2 Data analysis

Shape	2470 rows × 12 columns	
Summary	Consists of mean, STD, min, 25%, 50%, 75%, max	
No. of rows dropped	189 rows	
	Column type	No. of null values
S. No	float64	0
Year	float64	0
Month	float64	0
Day	Object	0
River name	Object	0
Location	Object	0
DO (mg/L)	float64	289
BOD (mg/L)	float64	84
COD (mg/L)	float64	1799
Total coliform (MPN/100 ml)	float64	344
Fecal coliform (MPN/100 ml)	float64	624
Category	Object	0

Table 2 summarizes the data analysis outcome.

Data visualization

Visualization is a statistical approach to data set analysis to summarize their main characteristics, by using statistical representations and other visualization methods. So, to analyze the data sets based on their main features, different line and bar plots were generated.

A model for analysis of water quality was created post data analysis. The model used in the presented work is wavelet genetic programming (WGP) model. WGP links wavelet analysis to the concept of genetic programming to analyze the DO values, to study the effectiveness of the model for water quality assessment.

Firstly, the length of an object must be found. For the presented work for all the rivers under consideration, based on their name from the River Name column using the *len()* function was checked.

Secondly, a new dataset was created for each river under consideration.

Thirdly, CWT was used as it plays a key role in the analysis of time–frequency information and helps in the building of classical short-time Fourier trans-

Fig. 3 Index map of the study area highlighting the sample points for the different rivers present in the UP region



form. In this work, the Morlet wavelet was selected as the mother wavelet as it helps in producing more accurate results than the Fourier transform.

Finally, genetic programming was used as a potential tool to develop efficient and simple functional relationships between hydrological variables. In this work, we assigned the genetic programming control parameters such as the number of generations.

Lastly, the model's performance was evaluated using accuracy metric. Accuracy is a ratio of correctly classified points to the total number of predictions. The range of accuracy is always between 0 and 1.

Study area and dataset used

This study was conducted on the rivers flowing in the state of Uttar Pradesh, India, between 23°52' N and 31°28' N latitudes and 77°3' E and 84°39' E longitudes. The network of around 32 sampling stations (Fig. 3) has been considered based on the availability

of data on different water quality parameters collected from 13 rivers in Uttar Pradesh. The sampling locations are spread across the whole state covering cities located in the western (like Noida, Ghaziabad, Meerut, and Bulandshahar), central (like Lucknow, Unnao, Sitapur, and Kanpur), and eastern (like Gorakhpur, Varanasi, Prayagraj, and Ghazipur) parts of the state.

For this work, data was taken from the UP-Pollution Control Board website titled "Water Quality Data of the Polluted River Stretches" (<http://www.uppccb.com/water-quality-data-stretches.htm>) for the years 2020 and 2021. The dataset used in this study consists of 2280 monthly river water samples that are collected between January 2020 and December 2021 from various sampling locations of the 13 different rivers present in UP. The collected samples of the river water were analyzed for their chemical and biological parameters such as DO, BOD, COD, total coliform, and fecal coliform to study the effectiveness of WGP model. Table 3 presents the names of the 13 rivers under consideration along with their sampling locations.

Table 3 Thirteen selected rivers along with their sampling locations

S. No	River	City	Sampling locations
1	Ami	Gorakhpur	U/s, before the confluence with Sariya Nala, Near Village Bharsarh, Sahjanwa; D/s, GIDA industrial effluent through Sariya Nala, Near Chhatai road Bridge Khajani; D/s near Lko, Vns road bridge Kauriram
2	Betwa	Banda	Hamirpur
3	Ganga	Kannauj	U/s and D/s Kannauj
		Kanpur	Bithoor; Bhairao Ghat; D/s Shuklaganj, Gola Ghat; Jajmau Bridge
		Raibareli	Dalmau
		Pratapgarh	Kala Kankar
		Kaushambi	Kada Ghat
		Prayagraj	U/s (Rasoolabad Ghat) and D/s Prayagraj
		Son Barsa	A/c Tamsa river, Sirsa
		Mirzapur	U/s (Vindhyachal) and D/s Mirzapur; Chunnar Pontoon Bridge
		Varanasi	U/s and D/s Varanasi
		Ghazipur	Tarighat D/s Ghazipur
		Bhausala	A/c Gomti River, Bhausala
4	Ghaghra	Gorakhpur	U/S Badhalganj
		Deoria	D/s Turtipur
5	Gomti	Sitapur	Dadhnamau Ghat, Sitapur
		Lucknow	Manjhighat; U/s Water Intake, Gaughan; Kudiyaighat; D/s Mohan Meakins; Nishatganj Bridge; U/s Bairaj; D/s Pipraghat; D/s After meeting of STP, Nala Bharwara
		Jaunpur	D/s Gomti
		Varanasi	Gomti before meeting river Ganga, Rajwari
6	Hindon	Saharanpur	D/s Maheshpur
		Meerut	Sardhana-Budhana Road, Vill Baparsi; Meerut Baghpat Road
		Ghaziabad	Karheda Village; Mohan Nagar Road Bridge; Chijarsi Bridge
		Noida	D/s Kulsera Bridge
7	Kali	Meerut	D/s M/s New Bananza, Paper mills vill., Saini-Mawana Road; Garh Road; Kharkhoda-Prikshit Garh Road
		Ghaziabad	Babugarh
		Bulandshahar	U/s (Devipura) and D/s (Mohan Kuteer)
		Aligarh	U/s (Before wave distillery) and D/s (After wave distillery), Ramghat Road, Atrauli
		Kasganj	Nadrai Gate
		Kannauj	U/s Kannauj
8	Ramganga	Moradabad	U/s Ramganga near Agwanpur; Moradabad-Rampur road bridge
		Rampur	D/s Ramganga, Shahabad
		Bareilly	U/s Ramganga Kapurpur village, Meerganj
		Shahjahanpur	D/s Ramganga FBD road bridge
		Farrukhabad	Ramganga
9	Rapti	Gorakhpur	Domingarh; D/s Gorakhpur
10	Sai	Unnao	Bani
		Raibareli	Sirsaghat; Rajghat; Picnic Spot; Gaighat
		Pratapgarh	Belami Ghat; D/s Stream Pratapgarh, Near Railway Bridge
		Jaunpur	Sai Nadi, Jabalpur
11	Saryu	Ayodhya	Ayodhya
12	Varuna	Varanasi	Rameshwar; Before meeting Ganga River

Table 3 (continued)

S. No	River	City	Sampling locations
13	Yamuna	Noida	U/s Okhla Barrage; D/s Village Gharbara/ Tilwara
		Vrindavan	U/s and D/s Vrindavan; Kesi Ghat
		Mathura	U/s and D/s Mathura; Shahpur; Vishram Ghat
		Agra	U/s Kailashghat; U/s Waterworks; D/s Rajmahal
		Firozabad	U/s and D/s Firozabad
		Etawah	U/s and D/s Etawah
		Allahabad	U/s Water Intake
		Prayagraj	D/s Balua Ghat; D/s Chhachar nala; D/s Emergency Outfall

Results and discussion

Data visualization

To analyze the data sets, the below mentioned plots were generated (Fig. 4 and Fig. 5):

The average DO values of 13 different rivers
 The average BOD values of 13 different rivers
 The average COD values of 13 different rivers
 The average total coliform (MPN/100 ml) values of 13 different rivers
 The average fecal coliform (MPN/100 ml) values of 13 different rivers
 How much-dissolved oxygen was predicted per month?
 Minimum and maximum. DO levels according to mean DO from January to December
 Minimum and maximum BOD according to mean BOD from January to December
 Minimum and maximum COD according to mean COD from January to December
 Minimum and maximum total coliform according to the mean total coliform from January to December
 Figure 4.1 to 4.5 represent the average DO values, BOD values, COD values, total coliform values and fecal coliform values for the 13 rivers under consideration, respectively. The average values have been computed over the values recorded for each month in the period from January 2020 to December 2021 for all sampling locations. Saryu River had the highest average DO value (Fig. 4.1), followed by Ganga, Ghaghra, Rapti, and Ramganga.
 Kali (East) river had the highest average BOD value (Fig. 4.2), followed by Hindon, Ami,

Yamuna, and Gomti. Critical reviews gathered from several studies have revealed that the increase in values of BOD is caused due to increase in TSS and temperature values.

Hindon river had the highest average COD value (Fig. 4.3), followed by Kali (East), Ami, Yamuna, and Gomti. Critical reviews gathered from several studies have revealed that the increase in values of COD is caused due to increase in nitrate, pH, and polluted water.

Kali (east) river had the highest average total coliform (Fig. 4.4) and fecal coliform (Fig. 4.5) values, followed by Hindon, Gomti, Rapti, and Varuna. Critical reviews gathered from several studies have revealed that the coliform levels of water increased due to poor sanitation and unhygienic condition of water.

Figure 5.1 and 5.2 represent the maximum and minimum DO values as per the mean DO, respectively. Figure 5.3 and 5.4 represent the maximum and minimum BOD values as per the mean BOD, respectively. Figure 5.5 and 5.6 represent the maximum and minimum COD values as per the mean COD, respectively. Figure 5.7 and 5.8 represent the maximum and minimum total coliform values as per the mean total coliform, respectively. The mean values for DO, BOD, COD, and total coliform have been computed for the time period from January 2020 to December 2021.

Based on the data visualization, the following observations were made:

1. Dissolved oxygen level was the highest in the month of February (Fig. 4.6), followed by December, March, May, and April.

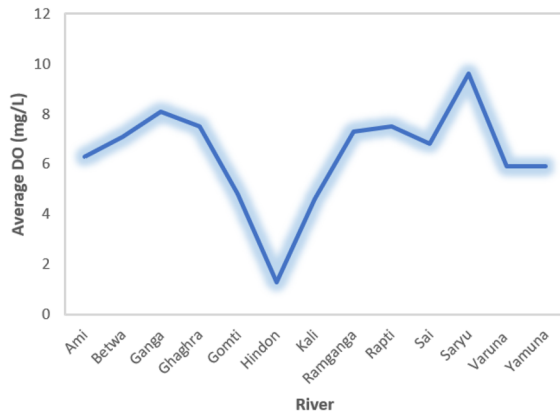


Fig 4.1 Average DO values

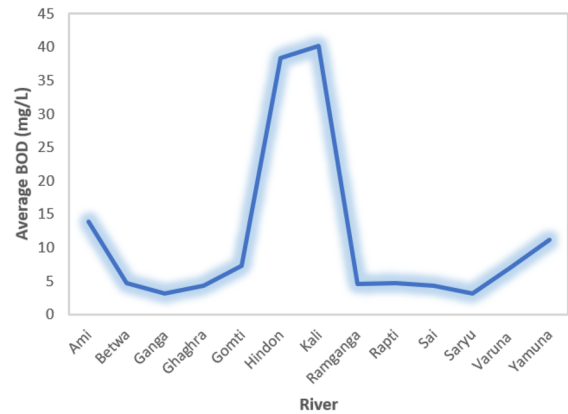


Fig 4.2 Average BOD values

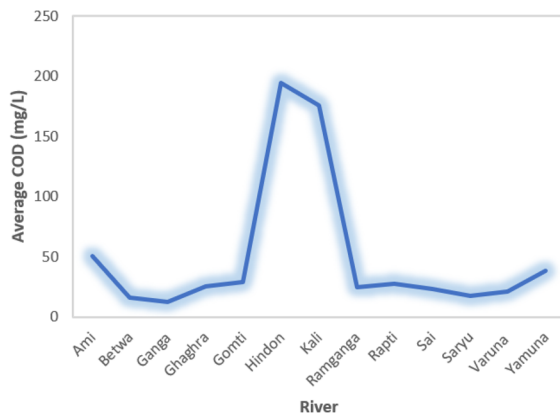


Fig 4.3 Average COD values

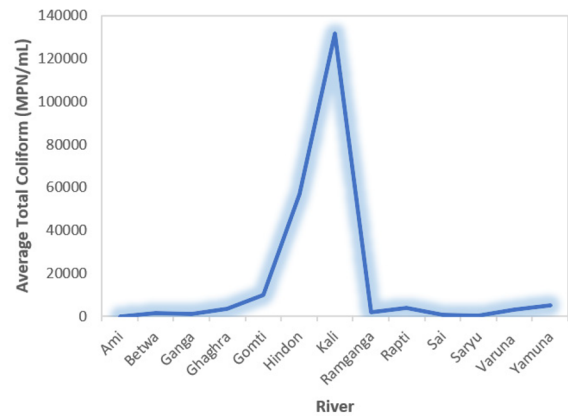


Fig 4.4 Average Total Coliform values

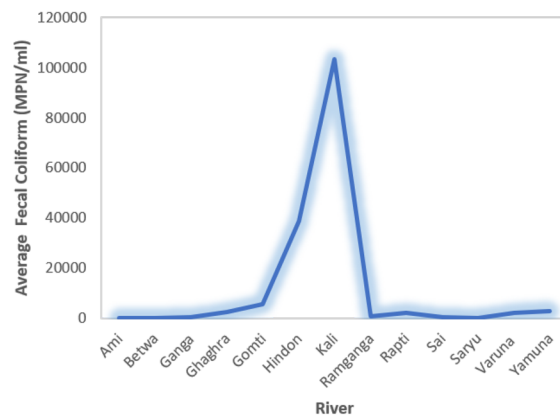


Fig 4.5 Average Fecal Coliform values

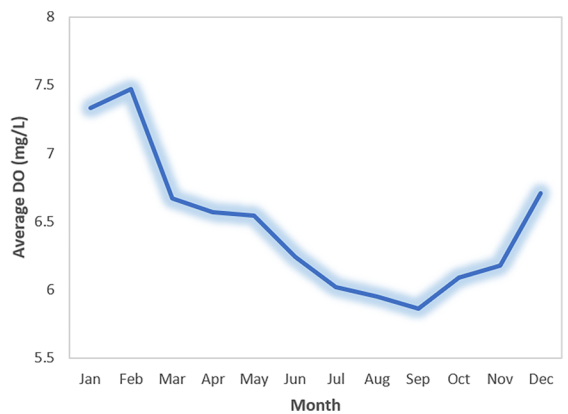


Fig 4.6 Dissolved Oxygen per Month

Fig. 4 4.1, 4.2, 4.3, 4.4, 4.5 and 4.6 represent the average DO values, BOD values, COD values, total coliform values, fecal coliform and dissolved oxygen per month values for the 13 rivers under consideration, respectively. The average values have

been computed over the values recorded for each month in the period from January 2020 to December 2021 for all sampling locations. Saryu river had the highest average DO value, followed by Ganga, Ghaghra, Rapti, and Ramganga

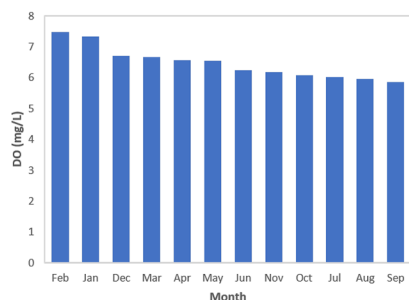


Fig 5.1 Max DO values according to mean DO from Jan-Dec

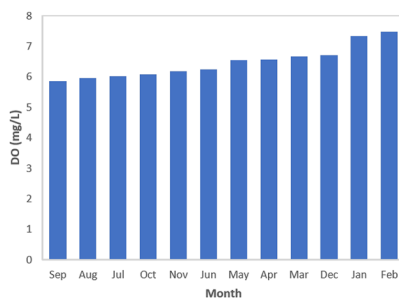


Fig 5.2 Min DO values according to mean DO from Jan-Dec

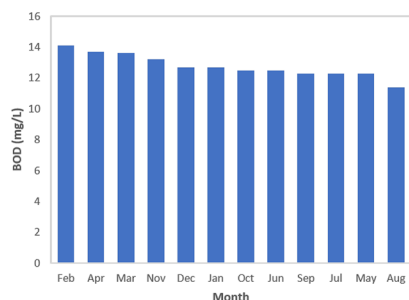


Fig 5.3 Max BOD values according to mean BOD from Jan-Dec

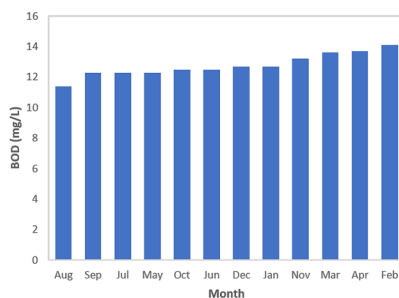


Fig 5.4 Min BOD values according to mean BOD from Jan-Dec

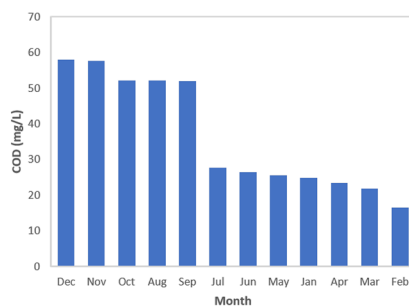


Fig 5.5 Max COD values according to mean COD from Jan-Dec

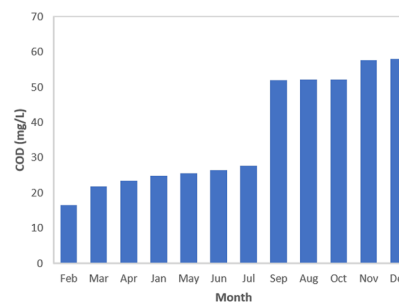


Fig 5.6 Min COD values according to mean COD from Jan-Dec

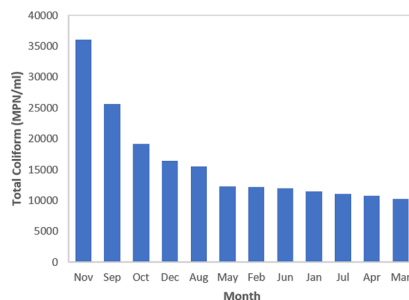


Fig 5.7 Max Total Coliform values according to mean Total coliform from Jan-Dec

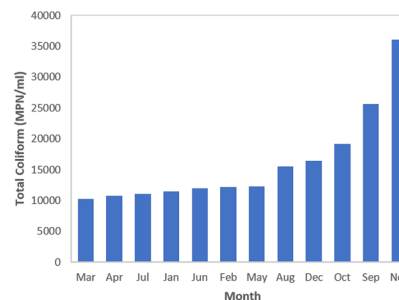


Fig 5.8 Min Total Coliform values according to mean Total coliform from Jan-Dec

Fig. 5 5.1 and 5.2 represent the maximum and minimum DO values as per the mean DO, respectively. 5.3 and 5.4 represent the maximum and minimum BOD values as per the mean BOD, respectively. 5.5 and 5.6 represent the maximum and minimum COD values as per the mean COD, respectively. 5.7

and 5.8 represent the maximum and minimum total coliform values as per the mean total coliform, respectively. The mean values for DO, BOD, COD, and total coliform have been computed for the time period from January 2020 to December 2021

2. The highest DO value according to mean DO was found to be for the month of February (Fig. 5.1), followed by January, December, March, and April. The lowest DO value according to the mean DO was found to be for the month of September (Fig. 5.2), followed by August and July.
3. The maximum value of BOD according to the mean BOD was observed for the month of February (Fig. 5.3), followed by the months of April, March, November, and January. The minimum BOD level according to the mean BOD was observed for the month of August (Fig. 5.4), followed by May and July.
4. COD value according to the mean COD was the highest during December (Fig. 5.5), followed by November, October, August, and September. COD value according to the mean COD was lowest during February (Fig. 5.6), followed by March, April, January, and May.
5. The highest total coliform value according to the mean total coliform was found to be for the month of November (Fig. 5.7), followed by September, October, December, and August. The lowest total coliform value according to the mean total coliform was found to be for the month of March (Fig. 5.8), followed by January, April, and July. For the months when total coliform was high, water was not safe for use, whereas in the months where total coliform was lower, the use of water was safe.

Significance of the findings

The work presented in the article has performed an extensive exploratory study on the sample data prior to model generation. The exploratory study has computed the average, minimum, and maximum values of DO, BOD, COD, total coliform, and fecal coliform from 13 different rivers from Uttar Pradesh. The average values have been computed over the values recorded for each month in the period from January 2020 to December 2021 for all sampling locations. It has been found from the

results that Kali River is one of the most contaminated river which is supported by its high levels of COD, BOD, total coliform, and fecal coliform. This fact is evident in the published reports and articles. Critical reviews gathered from several studies have also revealed that increase in values of DO is caused due to increase in temperature and salinity values. This has been shown from the results that DO levels increase from warmer to hotter months. Published research has also revealed that an increase in TSS values results in higher BOD values indicating that water is unfit for use. This has been shown from the fact that BOD levels have shown sharp increase from warmer to hotter months. Several studies have also revealed that with the decrease in the organic material concentrations, the COD value decreases. This has been shown from the results that COD levels are higher from December to September and lower in warmer months. This is due to fact of increased solubility in hotter months. The highest total coliform value was found to be for the month of November to August. As temperature rises, its value reduces in hotter months. In high temperature, the composition rate is more in warmer months. Thus, we can claim that our results of study are correlated with the published results.

Performance evaluation through accuracy

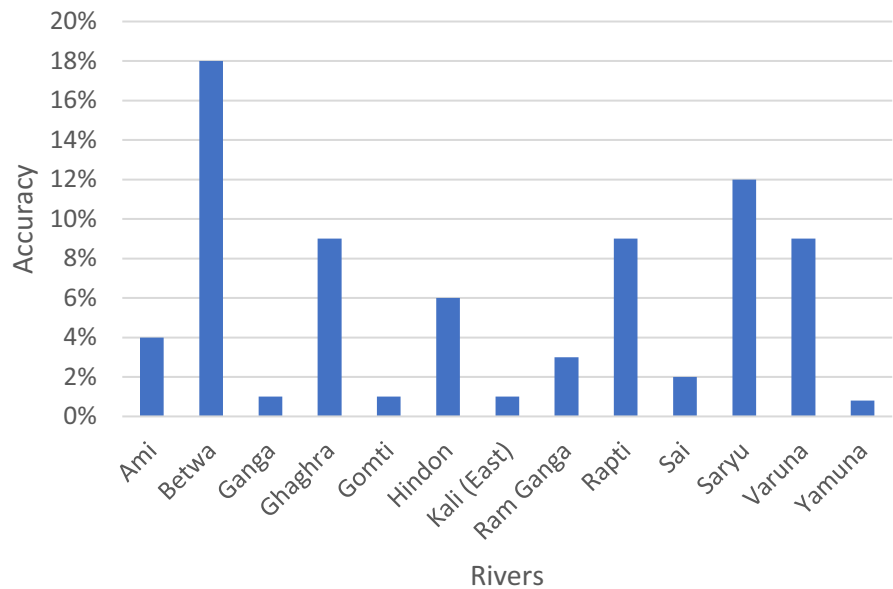
To evaluate the performance of the WGP model for assessing the water quality, its prediction accuracy was measured for the 13 rivers under consideration. Table 4 represents the different accuracies that were obtained using the WGP model for all the 13 different rivers present in the UP region. Figure 6 depicts the accuracies as a bar graph.

Therefore, by doing all the data collection, analysis, visualization, pre-processing, missing value imputations, model creation, and performance evaluation for the WGP model, it can be concluded that the WGP model proved efficient and effective for performing UP river water quality assessment based on DO parameter analysis.

Table 4 Accuracy metrics of 13 different rivers

River	Ami	Betwa	Ganga	Ghaghra	Gomti	Hindon	Kali (east)	Ram Ganga	Rapti	Sai	Saryu	Varuna	Yamuna
Accuracy	4%	18%	1%	9%	1%	6%	1%	3%	9%	2%	12%	9%	1%

Fig. 6 WGP model accuracies obtained for the different rivers present in the UP region



Conclusion

Water is one of the essential elements for existence on any planet. But with increasing pollution levels in different sources of water, the quality of water has been continuously deteriorating day by day. To check and control the contamination of surface and ground as well as river water, there is a strong need to work and produce promising results in this direction. This work is an attempt to contribute towards monitoring of water quality in the Indian rivers. We have empirically shown that dissolved oxygen (DO) is one of the most important parameters to assess water quality as it is considered a key determinant of pollution. We have categorically identified and visualized the results to show that the level of DO plays an important role as one of the key components responsible for deteriorating the quality of water. A wavelet genetic programming (WGP)-based workflow model has been built for the assessment of water quality in 13 rivers of Uttar Pradesh region. The model has been found capable to handle irrelevant and missing data in the training samples by automatically discarding the redundant and irrelevant data values from the source data. The

proposed WGP model is the first one, to the best of our knowledge that have been used for the study of water quality assessment in the Indian context. The use of Morlet wavelet in place of Db wavelet, as the mother wavelet as well as the MICE technique, used in imputing the missing values, has added to the success in showing the feasibility of using DO values to assess the water quality.

Author contribution Mansi Gaonkar: data curation, investigation and analysis, writing—original draft, and visualization. Bhawna Saxena: writing—review and editing. Sandeep Kumar Singh: writing—review and editing, and supervision.

Data availability River water data was taken from the UP-Pollution Control Board website titled “Water Quality Data of the Polluted River Stretches” available at <http://www.uppcb.com/water-quality-data-stretches.htm> for the years 2020 and 2021.

Declarations

Ethics approval All authors have read, understood, and have complied as applicable with the statement on “Ethical responsibilities of Authors” as found in the Instructions for Authors and are aware that with minor exceptions, no changes can be made to authorship once the paper is submitted.

Competing interests The authors declare no competing interests.

Appendix

S. No	Abbreviation	Full form
1	W-MGGP	Wavelet pre-processing with multi-gene genetic programming
2	WGEF	Wavelet gene expression programming
3	DWT	Discrete wavelet transform
4	LPM	Linear probability model
5	MGGP	Multi-gene genetic programming
6	PSR	Predictive state representation
7	FNN	False nearest neighbor
8	LASSO	Least absolute shrinkage and selection operator
9	GEP	Genetic evolutionary programming
10	FLGGP	Fixed-length gene genetic programming
11	LGP	Linear genetic programming
12	GP	Genetic programming
13	ARIMA	Autoregressive integrated moving average
14	WGP	Wavelet genetic programming
15	MSGP	Multi-stage genetic programming
16	GBT	Gradient boosting
17	ANN	Artificial neural networks
18	WANN	Wavelet artificial neural networks
19	ANFIS	Adaptive neuro-fuzzy inference system
20	WANFIS	Wavelet adaptive neuro-fuzzy inference system
21	FL	Fuzzy logic
22	SVM	Support vector machine
23	NF	Neuro-fuzzy
24	GA-NN	Genetic algorithm neural networks
25	WNN	Wavelet neural network
26	WSVR	Wavelet support vector regression
27	WLGP	Wavelet linear genetic programming
28	PCA	Principle component analysis
29	LSTM	Long short-term memory
30	RNNs	Recurrent neural networks
31	BP	Back propagation
32	SVR	Support vector regression
33	MSA	Multiple sequence alignments
34	MLP	Multilayer perceptron
35	MLR	Multiple linear regression
36	KNN	K-nearest neighbor
37	MODWT	Maximal overlap discrete wavelet transform
38	FWNN	Wavelet-based fuzzy neural network

S. No	Abbreviation	Full form
39	CWT	Continuous wavelet transform
40	DDM	Data-driven modeling
41	NAR	Non-linear autoregressive neural network
42	RF	Random forest
43	ABM	Agent-based modeling
44	GBU	Gradient boosting unit
45	EMSP	Effective maliciousness score of permission
46	GBT	Gradient boosting technique
47	OAM	Open application model
48	LEM	Learnable evolution model
49	EP	Evolutionary programming
50	AR	Augmented reality
51	WNF	Wavelet neuro-fuzzy
52	WTC	Wavelet task clustering
53	SARIMA	Seasonal autoregressive integrated moving average
54	FNN	Feedforward neural network
55	RMSE	Root mean square error
56	NSE	Nash–Sutcliffe efficiency
57	MAE	Mean absolute error
58	R ²	Coefficient of determination
59	MAPE	Mean absolute percentage error
60	MSE	Mean square error
61	AIC _c	Akaike information criterion
62	AARE	Average absolute relative error
63	MARE	Mean absolute relative error
64	AOI	Index of agreement
65	CE	Coefficient of efficiency
66	NSC	Nearest shrunken centroid
67	SSE	Sum of squared errors
68	MPE	Mean prediction error
69	R	Coefficient of correlation
70	RE	Relative error
71	MRSE	Mean root squared error
72	PARE	Pooled average relative underestimation and over estimation errors
73	NDEI	Non-dimensional error index
74	Ai	Accuracy factor
75	P	Pearson correlation coefficient
76	bias	Absolute value of the average forecast error
77	d	Willmott index of agreement
78	pMAPE	Mean absolute percentage errors of peaks
79	EVS	Explain variance score

S. No	Abbreviation	Full form
80	RPD	Residual prediction deviation
81	Dr	Discrepancy ratio
82	MSRE	Mean square relative error
83	MS4E	Mean higher order error
84	SI	Scatter index
85	MS	Mean squared error
86	SBC	Schwarz Bayesian criterion
87	DO	Dissolved oxygen
88	BOD	Biochemical oxygen demand
89	EC	Electrical conductivity
90	COD	Chemical oxygen demand
91	TDS	Total dissolved solids
92	TP	Total phosphorus
93	TKN	Total Kjeldahl nitrogen
94	NH ₃ -N	Ammoniacal nitrogen
95	RD	River depth
96	RW	River width
97	CA	Cultivated area
98	OA	Orchard area
99	SSL	Suspended sediment load
100	SSC	Suspended sediment concentration
101	NO ₃ ⁺	Nitrate
102	NH ₄ ⁺	Ammonium
103	PO ₄	Phosphate
104	TSS	Total suspended solids
105	NH ₄ ⁺ -N	Ammonium nitrogen
106	WT	Water temperature
107	TU	Turbidity
108	db	Daubechies4
109	dmey	Discrete Meyer
110	bior	Biorthogonal
111	coif	Coiflets
112	sym	Symlets (symmetrical wavelets)
113	MCAR	Missing completely at random
114	MAR	Missing at random
115	MNAR	Missing not at random
116	MICE	Multiple imputation by chained equations

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